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Illuminating Reservoir Dynamics: DAS VSP and Pre Survey Modeling for Enhanced EOR Monitoring, a Case Study from Carbonate Reservoir in UAE

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Abstract

Pre-survey modeling for Distributed Acoustic Sensing (DAS) Vertical Seismic Profiling (VSP) acquisition is a crucial step in the planning and execution of borehole seismic surveys. The primary objective of this study is to evaluate the effectiveness of DAS VSP in a horizontal well for monitoring fluid fronts, such as CO₂ and polymer, in an area under Simultaneous Injection of Miscible Gas and Polymer (SIMGAP). Pre-survey ray trace modeling was performed to assess the fiber optic cable's capability deployed in horizontal wells to detect 4D seismic signals prior to acquisition for optimizing reservoir monitoring and enhancing the understanding of subsurface fluid dynamics.

The pre-survey modeling demonstrated the importance of pre-planning prior to cable deployment. Different source line geometries were modeled. The scenario of using five parallel lines including a line along the polymer injector showed poor P-down and PP-up DAS VSP arrivals over the horizontal wellbore section due to the inherent limitations of DAS VSP in clearly capturing these arrivals in such geometries. Additionally, there is no significant value in having five parallel source lines, as it is not expected to image away from the horizontal polymer injector location. The PP-up and PS-up reflection moveouts are expected to be very similar over the horizontal wellbore, making processing and interpretation challenging. For perpendicular source lines to the fiber, the expected fiber sensitivity is worse for P- and S-wave modes, as the fiber angle is 90 degrees for all wave modes, resulting in reduced signal quality. Hence, the results indicate that the installed fiber is not optimal for achieving the study objectives using DAS VSP in the horizontal polymer injector. An alternative option to consider is to acquire a conventional 3D VSP survey in the vertical observer wells in the SINGAP area.

This study underscores the importance of pre-survey modeling prior to cable deployment in horizontal wells. DAS VSP surveys have huge potential in monitoring the reservoir behavior in EOR projects. However, proper planning via pre-survey modeling for determining the optimal survey geometry and cable deployment are keys to a successful time-lapse VSP study.

Introduction

Single well DAS VSP (Vargas et al., 2023) and multi-well DAS VSP (Mason et al., 2023) have been successfully deployed in Onshore and Offshore Abu Dhabi for subsurface imaging. In CO₂ Enhanced Oil Recovery (EOR) operations, 4D (time-lapse) seismic surveillance is a cornerstone for understanding reservoir dynamics (Mohamed et al., 2024) and optimizing injection strategies. Among the evolving technologies, DAS has emerged as a transformative tool, offering high-resolution, repeatable monitoring capabilities. Another successful implementation of DAS is being carried out in a carbonate saline aquifer CCS project in the UAE (Cambois et al., 2024; Mason et al., 2024; Kirstetter et al., 2025).

DAS has revolutionized the field of seismic data acquisition by providing continuous, high-resolution measurements along the length of a fiber optic cable. DAS VSP offers a powerful tool for subsurface imaging and monitoring, allowing real-time data acquisition, facilitating a better understanding of the reservoir dynamics.

Initially, the primary focus of fiber optic cable installation was geared towards Distributed Temperature Sensing (DTS) data acquisition. Consequently, the suitability of fiber optic cable for DAS VSP acquisition in horizontal well was not assessed. The success of a DAS VSP survey hinges significantly on the pre-survey modeling study. Spending the time to properly model and optimize the acquisition of the VSP is critical to minimizing cost and environmental footprint while maximizing the economic value of the VSP to the project (Schweigert and Schulte, 2019). This crucial stage employs simulations to forecast the subsurface seismic imaging coverage to various source and receiver configurations. The importance of pre-survey modeling cannot be overstated as these simulations help to elucidate the effect of different factors, including source and receiver type, spacing, and geometries on the subsurface coverage and data quality (Jie et al., 2018). By thoroughly conducting pre-survey modeling, the survey design can be meticulously tailored – if possible – to meet specific monitoring objectives. This iterative process involves defining geological targets, formulating hypotheses about subsurface conditions, constructing subsurface models, and testing various acquisition scenarios. Thereby reducing the risk of data quality issues and optimizing the cost.

Area Challenges

SIMGAP EOR project focuses on the part of the reservoir, where there is high permeability contrast between the upper and lower reservoirs. One of the key development challenges identified in the area is water over-ride and reverse coning. Water over-ride occurs when the injected water moves preferentially through the high permeability zones in the upper reservoir, bypassing the oil-rich areas in the lower reservoir. This results in inefficient oil recovery and increased water production. Reverse coning, on the other hand, refers to the downward movement of water/gas around producing wells leading to bypassed oil in the lower reservoir. Hence, results in inefficient oil recovery and poses a significant challenge for the oil production from the reservoir in the area.

The primary objective of the SIMGAP project is to improve sweep efficiency in the lower reservoir through simultaneous polymer and CO₂ injections in the upper and lower reservoir respectively as shown in Fig. 1. By doing so, it leads to optimizing the oil recovery. The polymer injected in the upper high permeability zone gives several benefits:

1. It creates a high-pressure barrier, so that the injected CO₂ in the lower zone is contained in the lower zone as long as possible giving CO₂ enough time and opportunity to sweep oil of the lower zone and delay breakthrough of gas in the producers.
2. It doesn't crossflow to the lower zone because of the polymer filtration effect and residual resistance factor (RRF) in the low permeability dense layer which is located between Upper and Lower zones.
3. It moves slowly because of the increased viscosity of the water-polymer mixture and RRF, delaying the water breakthrough in the producers.

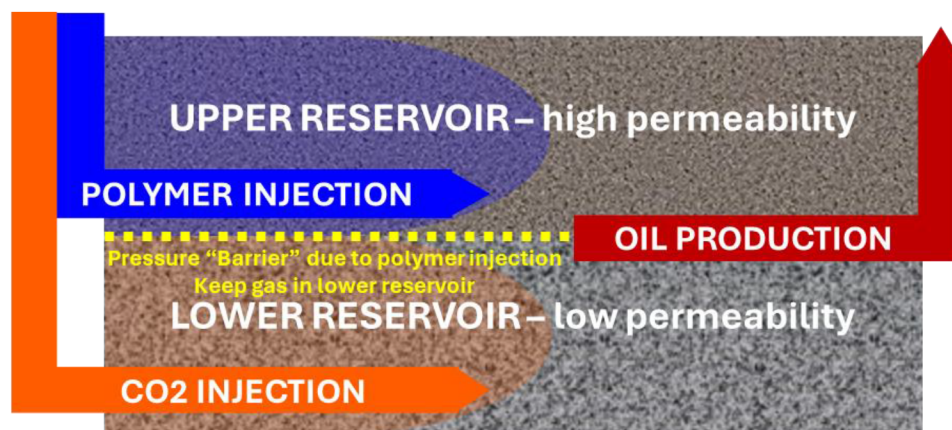


Figure 1—Illustration of SIMGAP EOR concept in the reservoir

The CO₂ injected in the lower low-permeability zone is miscible (through multi-contact miscibility mechanism) at the current reservoir pressure (which is going to be maintained in the future by injecting enough fluid). This will aid in enhancing the sweep efficiency of the oil and can reduce the residual oil saturation down to 4-5%.

Pilot Design and Fiber Optic Installation

SIMGAP EOR pilot features an injector-centered pilot geometry, which includes two central horizontal injectors, one is injecting CO₂ in the lower reservoir and the other is injecting polymer in the upper reservoir. Moreover, there are two flanking horizontal producers as shown in Fig. 2. The two central horizontal injectors are completed in the Upper reservoir and the Lower reservoir. The injectors are designed to inject 8 mmscf/d of CO₂ into the Lower reservoir and 4600 stb/d of polymer solution into the Upper reservoir with polymer concentration of about 1750 ppm. This dual injection placement aims to optimize the displacement of oil and improve overall reservoir recovery rates. The two flanking horizontal producers (PRD1 and PRD2) are also positioned in the Lower reservoir. These producers are designed to extract the swept oil, ensuring that the injected fluids effectively mobilize the trapped hydrocarbons. The horizontal length of these wells is approximately 3000 ft, with a well spacing of 250 meters. This configuration is intended to maximize the contact area with the reservoir and enhance the efficiency of the EOR process.

In addition to the injectors and producers, the pilot area includes two vertical observer wells (OBS). These wells are placed to monitor the performance of the pilot and gather valuable data such as fluid samples, pulse neutron & production logs on the reservoir's response to the injected fluids.

In terms of cable deployment, both the polymer injector and PRD1 have 11x11mm encapsulated hybrid cables which contains 18AWG conductor, a single-mode and multi-mode fibers. In both wells, the cable is clamped on the tubing. In the polymer injector, the fiber covers the entire horizontal well, from surface to the end of the horizontal section as shown dotted along the well in Fig. 2. Whereas PRD1's fiber was deployed along the vertical section of the well, it is worth noting that the primary objective of fiber optic cable installation was mainly to acquire DTS data in these wells.

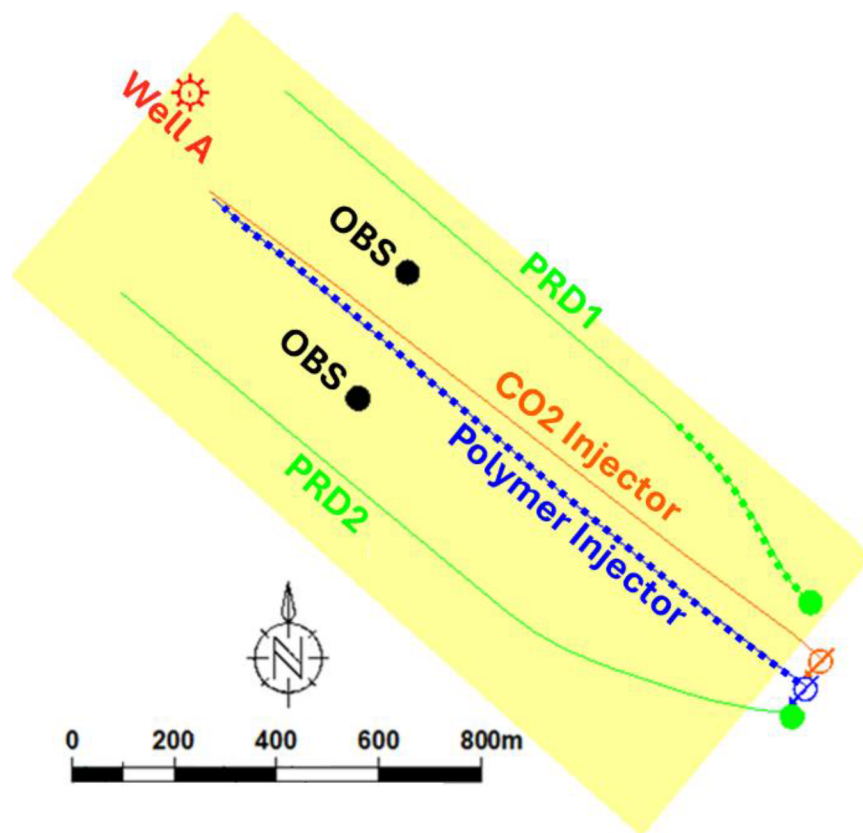


Figure 2—SIMGAP area with the pilot wells

Pre Survey Modeling

The feasibility of using DAS VSP for monitoring fluid fronts of injected polymer and CO₂ in the reservoir aims to assess the value of DAS VSP in reservoir monitoring by identifying unswept oil areas, quantifying sweep efficiency, and optimizing reservoir behavior.

The methodology for pre-survey modeling in DAS VSP acquisition involves several key steps. Initially, geological and geophysical data relevant to the survey area were collected and prepared to create a subsurface model that incorporates key geological formations. The pilot area in Fig. 2 shows the pilot wells' layout and the location of the nearby vertical well (Well A) which had VSP and log data available which were used to generate the velocity model in Fig. 3. The calibrated DTC log from Well A was blocked over major velocity contrasts including the overburden and reservoir as shown in Fig. 4a. Travel time residuals with zero-offset VSP data were observed to be in a reasonable range hence validating the model (Fig. 4b). For simplicity the subsurface model was considered isotropic with flat structure throughout the target area.

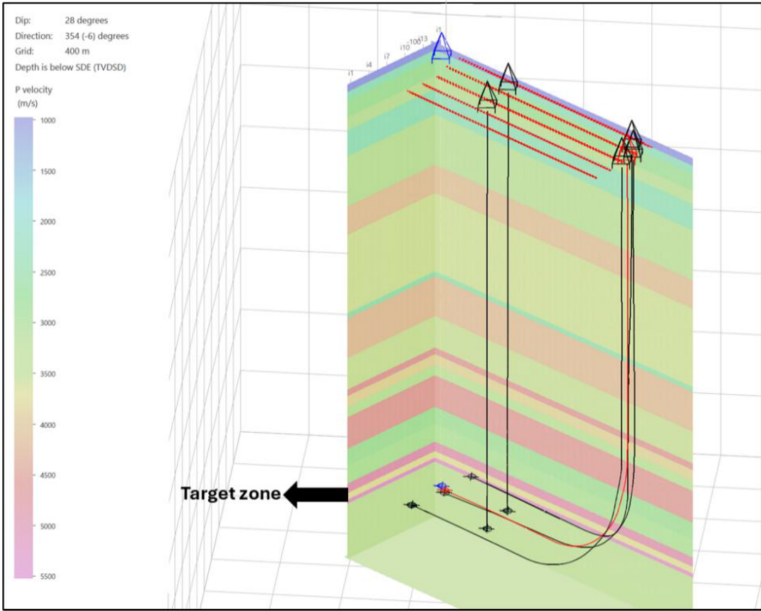


Figure 3—Velocity model of SIMGAP area with the pilot wells

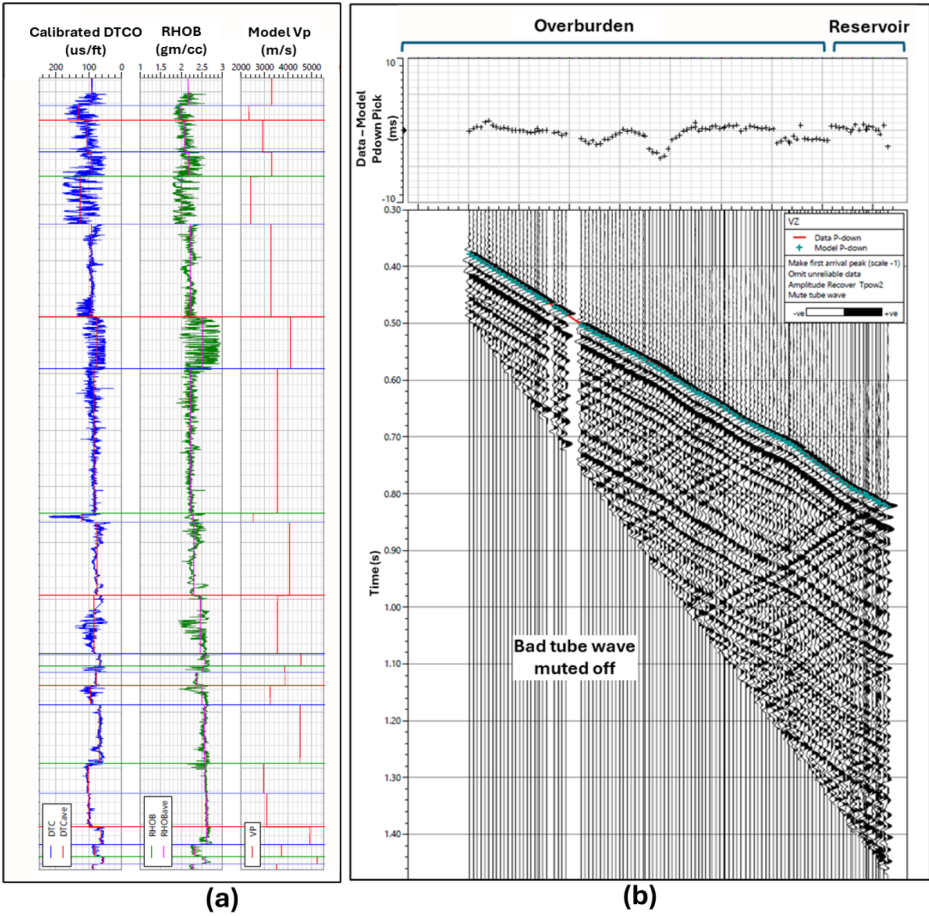


Figure 4—Velocity model from the nearby vertical well; (a) sonic, density logs and Vp model (b) zero-offset VSP data and travel time residuals

Survey design and modeling software was then used to simulate the seismic reflection coverage of the subsurface, testing different source configurations. The source layouts that were considered for feasibility in this study are the following:

- Scenario 1: Five parallel source lines to the Polymer Injector,
- Scenario 2: Three parallel source lines to the Polymer Injector + One perpendicular source line to the Polymer Injector,
- Scenario 3: One parallel source line to the Polymer Injector with large offset,
- Scenario 4: Sparse 200m source grid of conventional 3D VSP and DAS VSP.

Results

The analysis of the modeling results focused on determining the most effective survey geometry and evaluating the practicality of using DAS VSP in our scenario. The extent of subsurface coverage was carefully examined.

Before going into the results of the modeling, it is important to be aware of what arrivals can DAS VSP reliably record in horizontal wells. Fig. 5 shows the geophone and DAS response in a polar graph for P-wave and S-wave in fiber in horizontal well similar to figures in Baird et al. (2020). For good DAS VSP P-wave arrivals, it is essential to have arrival angles less than 45 degrees with the fiber. Conversely, for S-wave arrivals, PS-up energy may only be expected with reflection incidence angles greater than 10 degrees, and shear wave energy is more challenging to predict. It is also crucial to consider the reservoir interval that the recorded VSP arrivals will encounter, including P-down from above and PP-up and PS-up from the reservoir base which are reflectors coming from below.

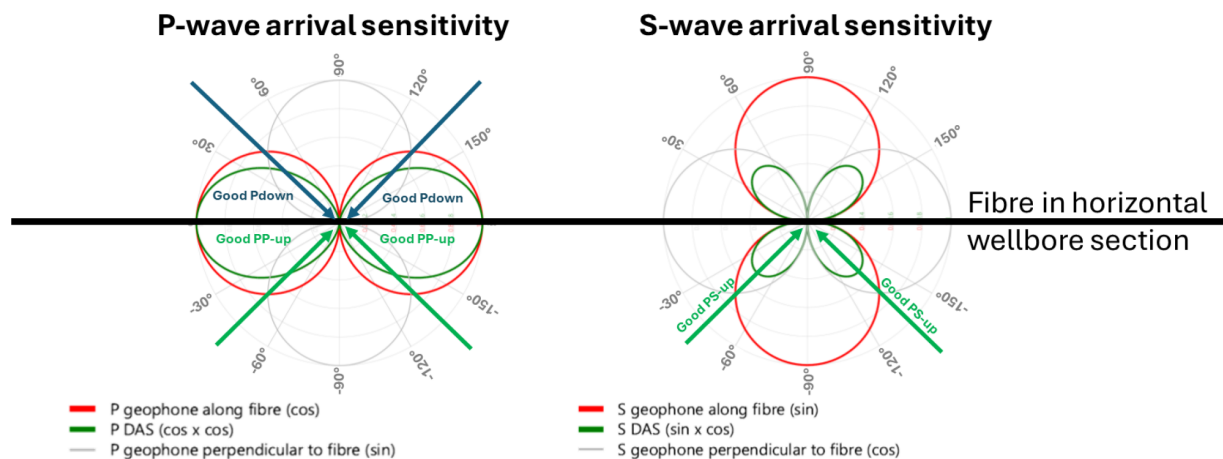


Figure 5—Illustration of arrivals recorded via DAS VSP installed in horizontal wells

In scenario 1 with five parallel source lines, the test started with the source line along the polymer injector at 4 different receivers as shown in Fig. 6. Overall, recording good P-down and PP-up DAS VSP arrivals over horizontal wellbore section is not expected since the arrivals are greater than 45 degrees (Fig. 6a).

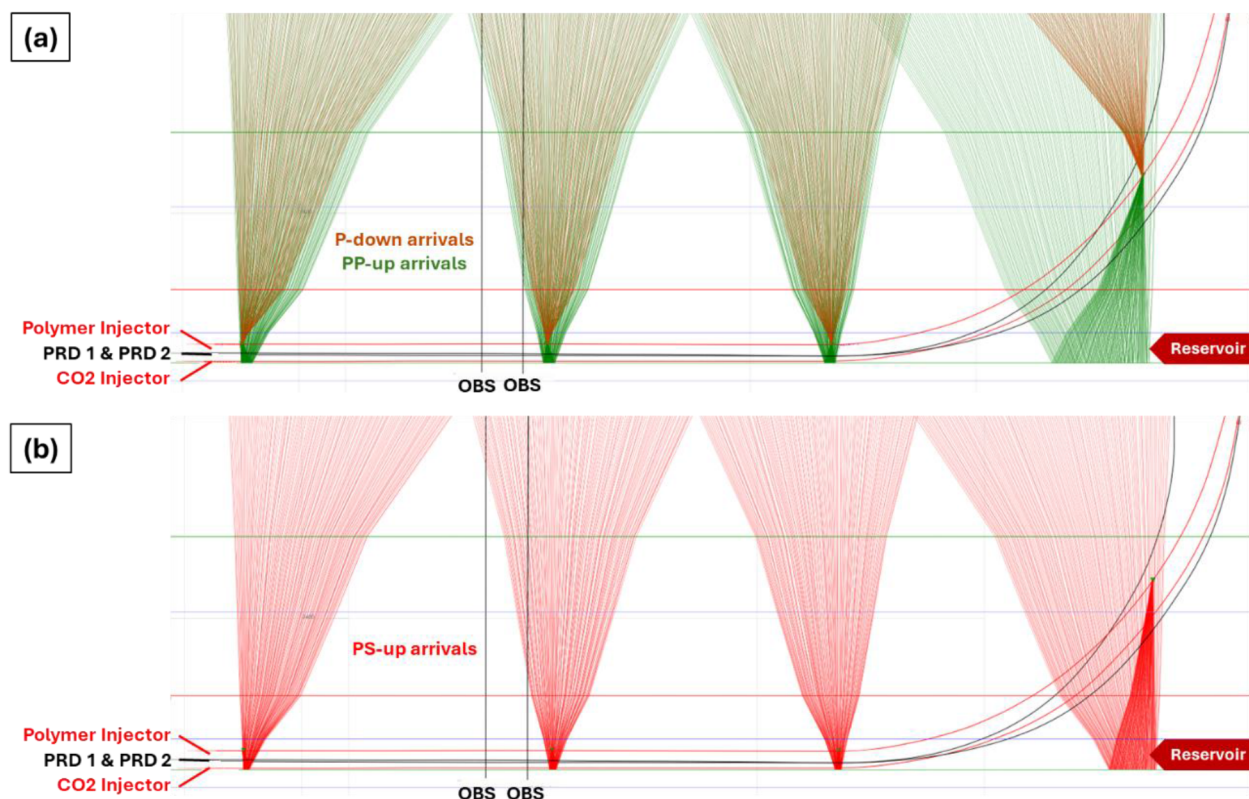


Figure 6—Cross section from the model showing DAS VSP P-down and PP-up in (a) and PS-up in (b) at 4 different receivers along the polymer injector

As for the PS-up arrivals, it could be recorded if generated as shown in Fig. 6b. Yet, since the objective of DAS VSP is monitoring fluid fronts, PS-up arrival is not expected to detect 4D signal in our scenario. Adding the additional parallel source lines show similar PS-up arrivals, hence there is no significant value in acquiring them as it is not expected to monitor away from the polymer injector location.

In addition, simple synthetic arrivals for all wave modes were generated for 4 shot locations shown in Fig. 7a. It was assumed that the fiber optic cable is well-coupled to the tubing, ensuring optimal signal transmission and minimal data attenuation and the synthetics are presumed to be free from any significant noise contamination, allowing for clearer interpretation of seismic arrivals. In Fig. 7b it is observed that over the horizontal wellbore section, PP-up and PS-up can hardly be differentiated, making it challenging for data processing and interpretation. Wang and Stewart (2024) highlighted challenges of data processing in horizontal well DAS VSP data.

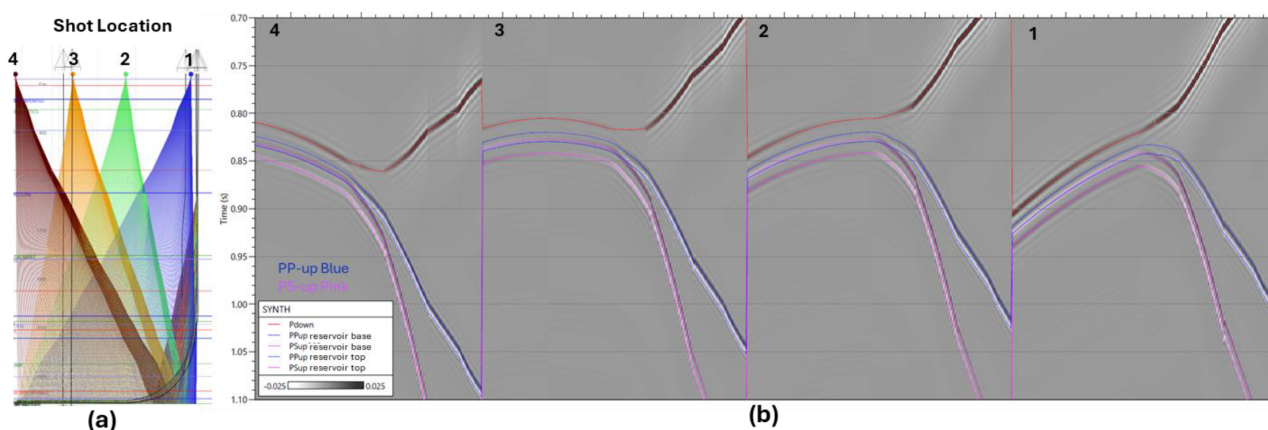


Figure 7—Simple synthetic arrivals of all wave modes for four selected shot locations

In the deviated build-up section, it is expected to have some P-wave energy coverage – assuming good coupling – as the angles of incident in this section are less than 45 degrees (refer to Fig. 5) however, this does not add value since it is away from the injection zone, see Fig. 8.

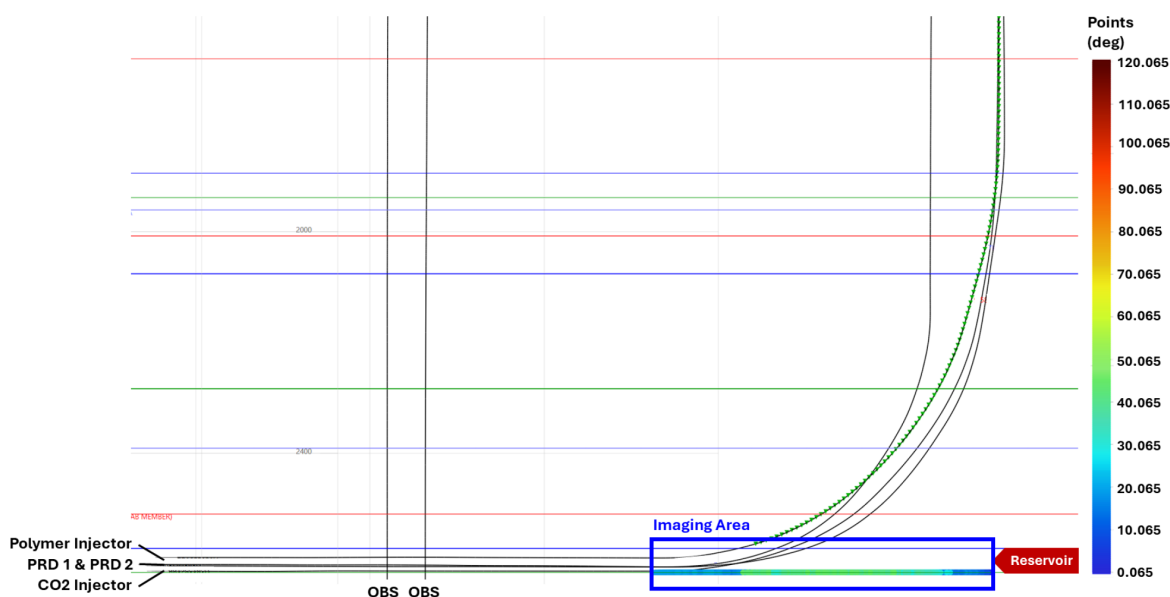


Figure 8—Cross section PP-up arrivals with less than 45 degrees are mainly recorded in the deviated section of the well

The three parallel source lines in scenario 2 resulted in similar findings to scenario 1. In addition to that, when testing the source line perpendicular to the polymer injector or fiber optic cable (Fig. 9), the expected fiber sensitivity is worse for P- and S-wave modes, as fiber angle is 90 degrees for all wave modes and according to the illustration in Fig. 5, this will not be useful for 4D monitoring.

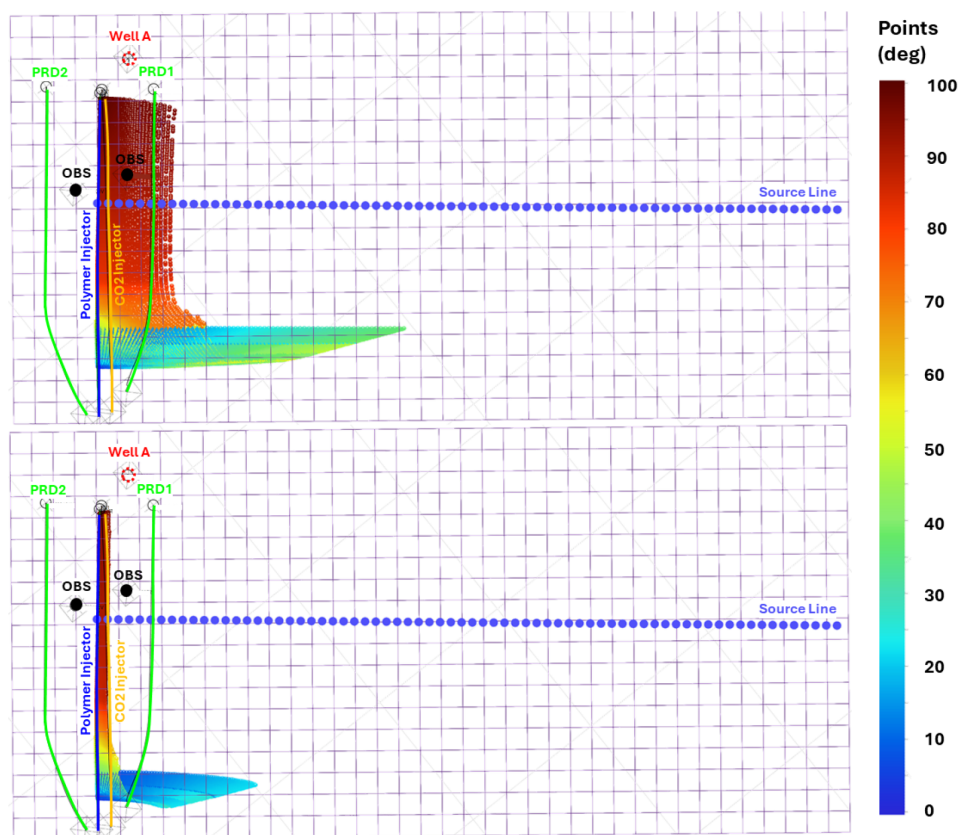


Figure 9—Map view of the pilot with a source line perpendicular to the fiber optic showing majority of arrivals along the horizontal section of the polymer injector reflect 90 degrees incident angle to the fibre

Extending the central source line ~2400 meters from the well head towards SE (i.e. opposite to the horizontal well direction) will enable better P-wave DAS VSP arrival angles over the horizontal wellbore section (Fig. 10). However, at the end of the horizontal section, the coverage becomes less, therefore does not fully achieve the monitoring objective across the pilot area.

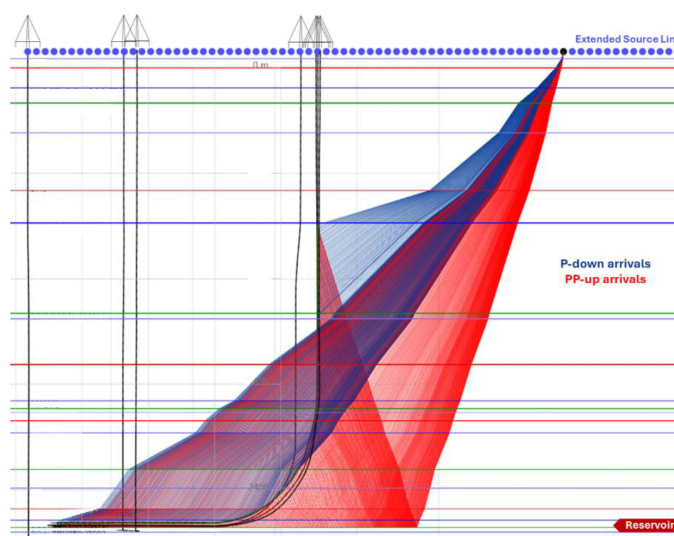


Figure 10—Cross section of the pilot with the extended source line showing P-down and PP-up arrivals

Given the inherent limitations of fiber optic geometry in horizontal wells, particularly the challenge of capturing meaningful DAS VSP arrivals, an alternative approach was explored in scenario 4. This involved

leveraging the observer wells (OBS) in the pilot area to acquire conventional 3D VSP surveys, thereby bypassing the angular sensitivity issues associated with horizontal DAS deployment. In this scenario, receivers are positioned from the end of the vertical well up to 1700 meters true vertical depth subsea (TVDSS), with a sparse 200-meter source grid and a source-receiver offset reaching up to 1700 meters. Fig. 11 illustrates the distribution of PP-up reflection points that were selected for incidence angles less than 35 degrees, optimizing the likelihood of achieving high-quality imaging.

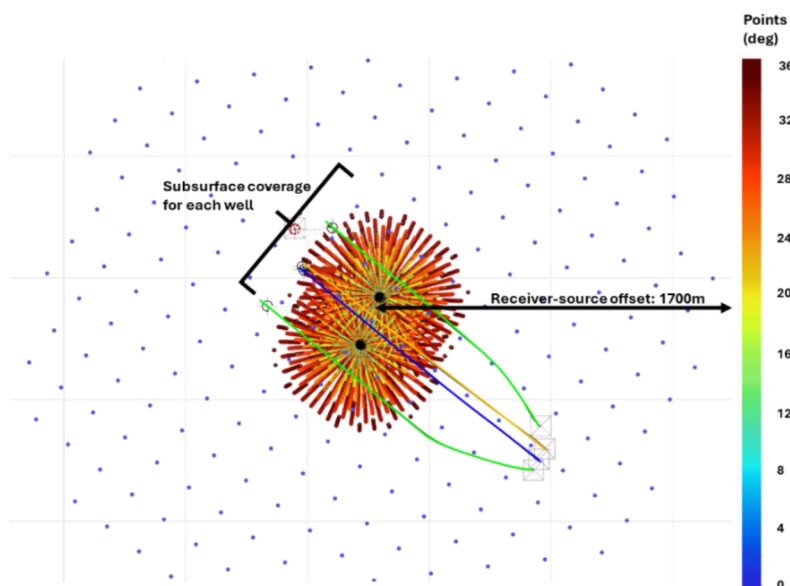


Figure 11—Map view of the pilot of the 3D VSP from the two observer wells showing the coverage of PP-up

With these assumptions and survey geometry, it becomes feasible to image a subsurface coverage radius of approximately 400 meters using just one observer well. While a single observer well can provide effective reservoir monitoring, dual-well VSP is recommended, as this approach yields comprehensive coverage of the pilot area. However, it is important to acknowledge that such acquisition plan would also entail higher costs compared to single-well monitoring.

Conclusion and Recommendations

In conclusion, pre-survey modeling plays a crucial role in advancing the planning and deployment of reservoir monitoring technologies, particularly those utilizing DAS with fiber optic cables. The pre survey modeling not only refines strategies for optimal cable placement and well selection, but it also directly informs how to maximize coverage of the target area while minimizing operational costs and inefficiencies. By simulating various survey geometries in advance, engineers and geophysicists can assess the sensitivity and effectiveness of different survey geometries. Ultimately, pre survey modeling helps to design monitoring programs that are best suited to the reservoir, ensuring that the data acquired will be enough to support confident decision-making throughout the life of the oilfield.

The results from pre-survey modeling highlight several insights regarding the challenges for effective reservoir monitoring using DAS VSP technology, in horizontal wells. One of the primary issues encountered is the difficulty in acquiring reliable DAS VSP recordings of P-wave arrivals that are essential for tracking fluid movement; this is especially problematic in horizontal well configurations. When source lines run parallel to the fiber optic cable installed along the horizontal section of the wellbore, the system predominantly captures S-wave arrivals, and the recording of P-waves becomes feasible only when the source is positioned at a significantly large offset—potentially around 2400 meters from the well head towards SE (i.e. opposite to the horizontal well direction). However, even in such cases, the system primarily

records high-angle P-wave reflections, which may not sufficiently represent fluid movement throughout the entire reservoir. Moreover, inline measurements that might be attempted along any other parallel line are unlikely to yield additional subsurface coverage, particularly in areas away from the injector. The situation becomes even more complex when considering source lines that run perpendicular to the horizontal fiber. In this configuration, all seismic arrivals are expected to intersect the fiber at a 90-degree angle, creating a scenario where neither P- nor S-wave arrivals are optimally detected by the fiber, thereby severely limiting the usability of DAS VSP in horizontal wells for imaging and 4D monitoring purposes.

To address these geometric and sensitivity limitations inherent in horizontal fiber deployments, the modeling also explored the scenario of using vertical observer wells for conventional 3D VSP surveys. The findings indicate that it is possible to achieve a subsurface monitoring radius of approximately 400 meters from a single well, capturing high-quality signal. Nevertheless, the dual-well VSP is recommended for optimal coverage, although at a higher cost.

Cambois et al. (2024) recommends installing the fiber behind casing instead of tubing, because of the strong tube wave and injection noise which cannot be fully removed from the data and will encounter issues for the up/down deconvolution. In addition, installing the fiber cemented behind the casing in a vertical wellbore improved signal to noise ratio. Mason et al. (2024) shows a test from a vertical well time-lapse DAS VSP CCS monitoring project in carbonate saline aquifer from onshore Abu Dhabi, inferring that in order to be able to detect changes in seismic response due to CO₂ injection and reduce the risk of misinterpretation, efforts should be made to place the DAS fiber as deep in the well as possible, as close to the injection zone as possible and preferably all the way through the injection zone.

This study signifies the challenges that might be encountered if DAS VSP is acquired in horizontal wells. In addition to that, it highlights the importance of pre-planning and pre-survey modeling prior to the deployment of fiber optic cables in any well as there are many factors which play a role in the success of an EOR monitoring project when using fiber optic cables, particularly for time-lapse DAS VSPs.

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